



Document Management System



Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Child Health

Code: CH-PAI-PRT-4-Vers.1.0

Effective Date: 16/12/2024

Target Review Date: 31/12/2027

Title: Diagnosis and Management of Children with Steven-Johnson Syndrome & Toxic Epidermal Necrolysis

Introduction:

Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN) are potentially life-threatening disorders and pediatric emergency characterized by widespread epidermal necrosis of skin and mucosa. Classification is based on the body surface area (BSA) affected, as SJS involving <10%, SJS-TEN 10–30% and TEN >30% of BSA. It can be complicated with infection and fluid loss leading to sepsis, hypovolemic shock, multiple organ failure, and death.

Frequency:

SJS/TEN is a rare disease, affecting 1 to 2 per million people each year. No current epidemiological details available for this disease in Omani children.

Clinical features:

- **Acute/short-term:**

- Clinical signs begin 7 to 21 days in the first exposure; however onset of symptoms can vary. In re exposure, the onset may happen earlier.
- Starts with unspecific symptoms such as fever, sore throat, stinging mucous membranes. There prodromal symptoms can precede cutaneous manifestations by a few days.
- Erythematous papules or purpuric macules with blisters.
- Oral mucosa, ocular and genital involvement in more than 90% of the patients with extensive and painful erosions which lead to lip crusts, odynophagia, photophobia, dysuria and painful evacuations.
- Water and electrolytic disorder, hypovolemia, renal failure, thermoregulatory unbalance and higher predisposition to sepsis.
- Multiple organ dysfunction.

- **Long-term effects:**

- Skin pigmentation
- Dryness of the skin and mucous membranes.
- Excessive sweating.

- Impaired taste.
- Difficult urination.
- Chronic dryness or inflammation of the eyes, which can lead to increased photophobia and vision impairment.

Causes:

Studies suggest that the *HLA-B* gene variations associated with SJS/TEN cause the immune system to react abnormally to certain medications. In a process that is not well understood, the drug triggers cytotoxic T cells and natural killer (NK) cells to release granulysin that destroys cells in the skin and mucous membranes. The death of these cells causes the blistering and peeling that is characteristic of SJS/TEN. Other HLA and non-HLA genes have also been studied as potential risk factors for SJS/TEN. Children with systemic lupus erythematosus or HIV infections are more susceptible to drug-induced SJS. In many cases, no definitive trigger for a children's SJS/TEN is ever discovered.

The most frequently triggers associated with SJS/TEN:

◦ **Drugs:**

Between 100 and 200 different drugs may be associated with SJS. Drugs discontinued more than 1 month prior to onset of mucocutaneous physical findings are highly unlikely to be a cause. SJS and TEN most often begin between 4 and 28 days after culprit drug administration.

Most common drugs:

- **Antibiotic:** vancomycin, levofloxacin, penicillins, sulfonamides, cephalosporins.
- **Anticonvulsants:** sodium valproate, phenytoin, oxcarbazepine, lamotrigine, phenobarbitone, phenytoin, ethosuximide, carbamazepine.
- **NSAIDs & analgesics:** diclofenac, ibuprofen, paracetamol
- **Others:** fluconazole, allopurinol, oseltamivir
- **Infections:**
 - **Viral infections:** various viruses are impacted including: HSV, CMV, EBV, HIV, coxsackievirus, influenza, hepatitis, mumps and enteroviruses.
 - **Bacterial infections:** Mycoplasma pneumonia, group A beta-hemolytic streptococci, diphtheria, brucellosis, lymphogranuloma venereum, mycobacteria, rickettsial infections, tularemia, and typhoid.
 - **Fungal infections:** coccidioidomycosis, dermatophytosis and histoplasmosis
 - **Protozoal:** Malaria and trichomoniasis.
- **Malignancies**
- **Unidentified**

Vaccination is NOT a cause of SJS/TEN

Differential Diagnoses:

- Erythema multiforme major.
- Staphylococcal scalded skin syndrome.
- Linear IgA bullous dermatosis.
- Bullous acute graft-versus-host disease.
- Bullous lupus erythematosus.
- Bullous pemphigoid.
- Epidermolysis bullosa acquisita.
- Kawasaki disease (early-stage erythema no blisters).
- Behçet disease.
- Generalized bullous fixed-drug eruption.
- Pemphigus vulgaris.
- Paraneoplastic pemphigus.

- **Initial assessment on presentation:**

Consider low threshold for admission to higher intensive care facility.

- Take a focused and detailed history.
- Exclude important possible differential diagnosis and pursue skin biopsy for diagnostic uncertainty.
- Investigate potential infectious etiology in all children & consider targeted antimicrobial therapy as appropriate.
- Record all medicines received over the preceding 2 months, including over the counter.
- Immediately discontinue any potential drug causing SJS & TEN.

ALDEN (ALgorithm of Drug causality in Epidermal Necrolysis) is an online tool that can be used to predict likely causality of a drug reaction.

- **Clinical examination:**

Strict hand hygiene before evaluation.

- Record vital signs and assess the need for respiratory support.
- Examine respiratory system to exclude pneumonia/respiratory compromise.
- Examine skin: look for target lesions, particularly atypical targets for erythema multiforme, purpuric macules, blisters, and areas of epidermal detachment.
- Nikolsky's sign.
- Examine mouth, eyes and genitalia (including perianal skin) looking for mucositis, blisters and erosions
- Record the extent of erythema and extent of epidermal involvement by estimating the percentage of involved body surface area (BSA).
- Protective isolation.

- **Consultation:**

Needed within 24 hours of diagnosis.

- Immunologist/pediatric allergist assessment.
- Dermatology assessment.
- Ophthalmology assessment.
- Others: MDT as indicated with infectious diseases, pain team, wound nurse, dietetics, pharmacy, ear, nose and throat team for further airway assessment.

- **Stabilization:**

- Airway, breathing and circulation assessment. Ensure immediate availability of appropriate equipment for a difficult intubation.
- Establish peripheral venous access; where possible, insert a peripheral cannula on a non-lesioned skin; commence appropriate intravenous fluid resuscitation if clinically indicated; beware of hyponatremia. Accurately record fluid intake and output and balance.
- Insert a nasogastric tube and institute nasogastric feeding if indicated.
- Insert a urinary catheter for accurate output monitoring and if urogenital involvement is causing significant dysuria or retention.

- **Investigations:**

- **Order the following set of investigations:**

- Full blood count, C-reactive protein, urea and electrolytes, liver function tests, glucose, magnesium, phosphate, bicarbonate, base excess, lactate and coagulation studies.
- Infection screening as clinically relevant (like mycoplasma serology/culture and HSV PCR, etc). Bacterial swabs from lesional skin for culture and sensitivity
- Chest X-ray.

- **General management:**

- Complete aseptic measures.
- Administer systemic antibiotics only if there are clinical signs of systemic infection. The choice of systemic antibiotic should be guided by local microbiological advice.
- Encourage early mobilization and limb physiotherapy.

- **Skin care:**

- This may involve a conservative and/or surgical:
 - Handle the skin carefully and reduce shearing, minimize the extent of epidermal detachment.
 - Minimize or avoid the use of sphygmomanometer cuffs, adhesive electrocardiogram (ECG) leads, adhesive dressings and identification wrist tags
 - Perform daily assessment of the extent of skin involvement and epidermal detachment
 - Skin swabs and aspiration of infected blisters for fluid microbial studies.
 - Regular cleansing of the wounds and intact skin by irrigating using warmed sterile water, saline or an antimicrobial agent.
 - Consider a greasy emollient, e.g. 50% white soft paraffin with 50% liquid paraffin (50/50 WSP/LP), applied over the whole skin, including denuded areas, every 2–4 h during the acute phase. Gauze dressing impregnated with paraffin can be applied.
 - Apply a topical antimicrobial agent like fusidic acid or mupirocin to sloughy areas.
 - Topical corticosteroid therapy to non-detached erythematous skin and mucosal areas, once infection has been excluded or treated.
- Surgical debridement may be needed if medical therapy fails to control the integrity of skin.

- **Mouth care:**

- Daily oral review during the acute phase.
- Apply white soft paraffin ointment to the lips or petroleum jelly every 2 h during the acute phase.
- Clean the mouth daily with warm saline mouthwashes or an oral sponge.
- Apply an anti-inflammatory oral rinse or spray every 2–4 h, particularly before eating.
- Corticosteroid cream or ointment applied topically to affected areas (including lips) during acute phase.

- **Eye care:**

- Initial examination should consider the extent of eyelid, conjunctival and corneal involvement.
- Daily ophthalmology review during the acute phase.
- Apply artificial tears every six hours during acute phase.

- **Immunomodulatory therapies:**

- The systemic treatment of SJS/TEN remains controversial.
- Treatments reported include systemic corticosteroids, intravenous immunoglobulins (IVIg), cyclosporine & plasmapheresis.
- In children and young people, systemic corticosteroids and IVIg are the two most used treatments. However, cyclosporine use is on the rise.

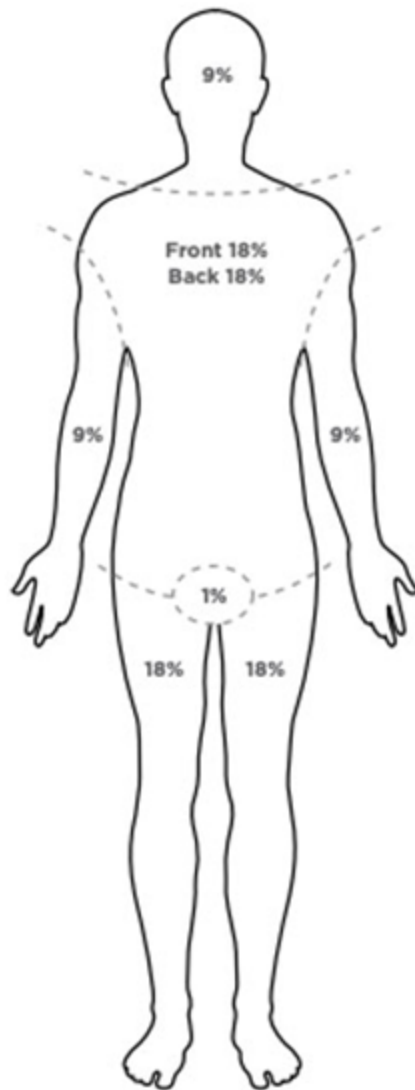
- **Systemic corticosteroids:**

- Individual case reports indicate that early administration of systemic glucocorticosteroids may limit disease progression and reduce morbidity and mortality.

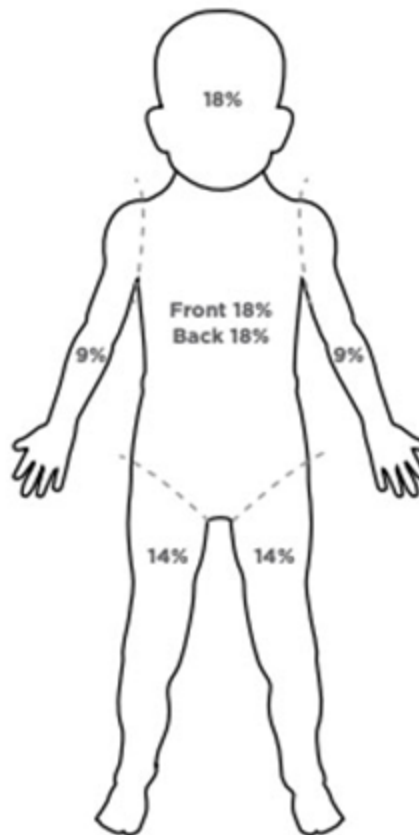
- Dexamethasone 1mg/kg for 7 days then to be tapered over 5 days after subsidence of erythema.
- Methylprednisolone 1-2 mg/kg/day over few days then followed by oral prednisolone 1 mg/kg/day in tapering doses (total upto 2 weeks)
- Methylprednisolone pulse 10 – 30 mg/kg over few days then followed by oral prednisolone 1 mg/kg/day in tapering doses. (total 2 weeks)
- Topical corticosteroid therapy
- **Intravenous immunoglobulins:**
 - Shorten lengths of hospital stay. Reduce mortality. Faster healing.
 - Doses: 0.25–1.5 g/kg daily; 1–5 days. On average IVIG often given at a dose of 1 gram/kg/day for 3 days.
- **Cyclosporine:**
 - Halting disease progression. Fostering re-epithelialization. Reducing mortality.
 - Dose: 3-5 mg/kg/d orally or intravenously for an average of 7 - 10 days. Half the dose can be given over the next week or so if indicated.
 - **There is a debate in the literature about the best strategy with most effective and safe mono- or poly- immunomodulatory therapy for SJS/TEN.** You can discuss with allergy and immunology team for more advice; however, a commoner practice is to start one of the following options:
 1. Systemic corticosteroids with IVIG
 2. Cyclosporin
- **Discharge and follow-up:**
 - Discuss potential long-term problems including skin pigmentation, skin scarring, nail dystrophy, eye, oral, dental, respiratory and urogenital complications.
 - Discuss the likely cause. If there are multiple potential causes, give balanced advice on the likely risk/benefit regarding re-exposure.
 - Discuss the risk of recurrence according to the likely cause.
 - For drug allergy, document it in the patient's notes, provide alert bracelet and provide parents with written information for drug avoidance.
 - Explain the treatment plan including duration and doses of corticosteroids or others and discuss the possible side effects for each medication.
 - Organize an outpatient clinic follow up within few weeks of discharge with pediatric immunologist.
- **Prognosis:**
 - Mortality rates in patients with TEN are approximately 25–35%, whereas mortality in SJS is estimated to be 5%.
 - In recent years, the use of the SCORTEN (SCORE of Toxic Epidermal Necrolysis) scale has been proposed to assess the severity of TEN cases and to predict the risk for death.
 - The scoring system is composed of seven features with identical weightings:
 - Age.
 - Heart rate.
 - Presence of malignancy.
 - Percentage of epidermal detachment at admission.
 - Serum urea level.
 - Bicarbonate level.
 - Serum glucose level.
 - The leading cause of death from TEN is multiple organ failure originating from sepsis. Other causes are gastrointestinal bleeding, pulmonary edema, pulmonary embolism and myocardial infarction.

Age > 40 years	1
Tachycardia > 120 bpm	1
Neoplasia	1
Initial detachment > 10%	1
Serum urea > 10 mmol/L	1
Serum bicarbonate < 20mmol/L	1
Blood glucose > 14 mmol/L	1

SCORTEN	Mortality (%)
0-1	3
2	12
3	35
4	58
≥ 5	90



Adult



**Paediatric
<9 years**



**Palmar, palm + fingers
of patient = 1%**

10.1016/j.jaip.2018.05.024.

- Liu W et al. A retrospective analysis of Stevens-Johnson syndrome/toxic epidermal necrolysis treated with corticosteroids. *Int J Dermatol.* 2016 Dec;55(12):1408-1413. doi: 10.1111/ijd.13379.
- McPherson T et al. British Association of Dermatologists' guidelines for the management of Stevens-Johnson syndrome/toxic epidermal necrolysis in children and young people, 2018. *Br J Dermatol.* 2019 Jul;181(1):37-54. doi: 10.1111/bjd.17841.
- Hsu DY et al. Pediatric Stevens-Johnson syndrome and toxic epidermal necrolysis in the United States. *J Am Acad Dermatol.* 2017 May;76(5):811-817.e4. doi: 10.1016/j.jaad.2016.12.024.
- KirchhofMGet al. Retrospective review of Stevens-Johnson syndrome/toxic epidermal necrolysis treatment comparing intravenous immunoglobulin with cyclosporine. *J Am Acad Dermatol.* 2014 Nov;71(5):941-7. doi: 10.1016/j.jaad.2014.07.016.

Written By: Dr.Khwater Abdul Rahman

Checked By: Dr.Nashat Al Sukaiti / Dr.Tariq Al Farsi

Authorized by: KHOULA JULENDA AL SAID

[Back](#)



© Copyright 2018 reserved 'Royal Hospital- Information
Technology Department'